Backpropagation from scratch

Friday, February 20, 2026 5:05 PM

Regression

Study hours	Sleep hours	H.S.C marks
2	3	3
5	6	4
7	8	15

L = 3

1

[2 2 1]

A[0] A[1] W[1] A[1] W[2] A[2]

2

x_1

b11

b11

$y^{\wedge}$

2X2

3

2 2 1

epoch = 5

for i in range(X.shape[0]):

→ Select 1 row (random)

→ predict (using forward prop)

→ Calculate loss (mse)

→ update w & b

$$\left[w_n = w_0 - \eta \frac{\partial L}{\partial w} \right]$$

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$

w^2_{11}

w^2_{21}

Auto

$$\frac{\partial L}{\partial w^2_{11}} = -2(y - \hat{y})\, o_{11}$$

$$\frac{\partial L}{\partial w^2_{21}} = -2(y - \hat{y})\, o_{12}$$

$$\frac{\partial L}{\partial b_{21}} = -2(y - \hat{y})$$

$$\frac{\partial L}{\partial w^1_{11}} = -2(y - \hat{y})\, w^2_{11} \cdot x^i_{11}$$

$$\frac{\partial L}{\partial w^1_{21}} = -2(y - \hat{y})\, w^2_{11} \times x_{12}$$

$$\frac{\partial L}{\partial b_{11}} = -2(y - \hat{y})\, w^2_{11}$$

$$\frac{\partial L}{\partial w^1_{12}} = -2(y - \hat{y})\, w^2_{21} \cdot x^i_{11}$$

$$\frac{\partial L}{\partial w^1_{22}} = -2(y - \hat{y})\, w^2_{21}\, x^i_{12}$$

$$\frac{\partial L}{\partial b_{12}} = -2(y - \hat{y})\, w^2_{21}$$

$$\frac{\partial L}{\partial w^2_{11}} \quad \frac{\partial L}{\partial w^2_{21}} \quad \frac{\partial L}{\partial b_{21}} \quad \Big| \quad \frac{\partial L}{\partial w^1_{11}} \quad \frac{\partial L}{\partial w^1_{21}} \quad \frac{\partial L}{\partial b_{11}} \quad \Big| \quad \frac{\partial L}{\partial w^1_{12}} \quad \frac{\partial L}{\partial w^1_{22}} \quad \frac{\partial L}{\partial b_{12}}$$

training pipeline

1) weight init → $\boxed{w = \pm \; b = 0}$
2) forward prop
3) weight update
4) loop